Beyond Binary Labels: Structured Multi-Task Supervision and Functional Robustness of Lightweight Hate Speech Detection

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Abstract

We propose HateLens, a structured multi-task extension of parameter-efficient fine-tuning (PEFT) of small decoder language models for hate speech detection. Building on Sen et al. [2024], who demonstrated that LoRA-tuned TinyLlama-1.1B achieves competitive binary accuracy on DynaHate and HateEval, we ask whether jointly predicting auxiliary attributes—hate type, target group, and explicitness—together with optional rationale-token supervision and pair-consistency regularization, is empirically associated with improved out-of-distribution robustness. We train on DynaHate (41k entries) with optional HateXplain joint data and evaluate on three benchmarks: in-domain test splits, cross-benchmark transfer to HateEval, and functional robustness via HateCheck (3,728 structured test cases). Compared to a controlled binary baseline (BIN-C), our best structured model (JOINT) improves HateCheck F1 by $+1.6$ pp ($0.905 \rightarrow 0.921$) and DynaHate F1 by $+1.3$ pp ($0.802 \rightarrow 0.815$), with only 780K trainable parameters. However, a post-review head-path control (BIN-FE): a `FEAT_EXTR` binary model with no auxiliary losses—achieves HateCheck F1 of 0.937, showing that the $+1.6$ pp gain is primarily architectural rather than a consequence of multi-task supervision. A separate data-isolation control (HX-NO-RAT) shows that HateXplain *data*—not the rationale loss—is the primary driver of HateEval cross-benchmark improvement (0.645 vs. Joint 0.615). Multi-seed evaluation (three seeds) confirms that HateCheck gains are within variance (± 1.7 pp for BIN-C), preventing strong causal claims. A controlled PEFT comparison (LoRA, QLoRA, DoRA) on a binary setup shows all three variants within 1 pp on HateCheck F1 (0.870–0.880), supporting the use of quantized adapters for deployment. Code and data loaders are publicly available.[1]

1 Introduction

Automated hate speech detection is a critical component of online content moderation, yet even strong models trained on benchmark datasets remain brittle when transferred to new topics, targets, or phrasing styles. A recurring finding across evaluations is that high held-out accuracy does not guarantee robustness on the edge cases that matter most in deployment: denial of hate (*"I'm just joking"*), references involving reclaimed slurs, or compositional rephrasing that preserves hateful intent while evading keyword filters [Röttger et al., 2021].

The gap in existing work. Sen et al. [2024] showed that LoRA fine-tuning of small decoder language models (TinyLlama, Phi-2, OPT-1.3B) achieves over 80% accuracy on DynaHate [Vidgen et al., 2021] and HateEval [Basile et al., 2019]— competitive with models ten times larger. Their work

[1] `https://github.com/EmaRimoldi/HateLens`

Preprint. Under review.

focuses exclusively on binary classification: is a post hateful or not? Binary supervision provides no signal about *why* a post should be classified as hateful, nor about the *type* or *target* of the hate. Two practical problems follow from this: (1) models that latch onto surface features—slur presence, identity terms—fail on adversarially generated edge cases of the kind HateCheck probes; (2) binary models trained on a single dataset transfer poorly to out-of-distribution benchmarks even when in-domain accuracy is high.

Our approach. We extend TinyLlama+LoRA with *structured multi-task supervision*: a shared PEFT backbone is joined by lightweight heads that predict the primary hate label, the hate type (e.g., derogation, animosity, dehumanization), the target group (e.g., race, gender), and the explicitness of expression. When HateXplain rationale annotations are available, we add token-level rationale supervision [Mathew et al., 2021]. When DynaHate contrastive pair metadata is available, we apply pair-consistency regularization. The backbone remains frozen except for LoRA adapters on the key/value projections, keeping trainable parameters at 780K.

Multi-task and multi-aspect hate detection has been studied before, largely with encoder models and shared tasks (see Section 2). Our contribution is narrower but practically motivated: we bring structured supervision into the tiny-decoder PEFT regime, quantify the controlled effect against a matched binary baseline, and evaluate robustness with HateCheck—a combination that has not been systematically reported in this setup.

Contributions.

- We report that structured multi-task supervision is associated with improved HateCheck robustness ($+1.6$ pp F1: $0.905 \rightarrow 0.921$, BIN-C vs. JOINT) and DynaHate in-domain F1 ($+1.3$ pp) over a controlled binary baseline using identical PEFT configuration and the same training data. The larger $+10.6$ pp gain vs. the legacy baseline (BIN-L) is partly attributable to training-configuration differences rather than structured objectives alone.
- We present ablation studies consistent with pair-consistency regularization improving cross-benchmark transfer to HateEval ($+2.1$ pp over STRUCT), and joint HateXplain training adding $+0.8$ pp HateCheck F1. Because controls are partially confounded (Section 6), we report these as empirical associations, not causal claims.
- We compare LoRA, QLoRA, and DoRA under controlled binary conditions, finding all three within 1 pp on HateCheck F1 and validating quantized adapters for deployment.
- We release HateLens, an open codebase with reproducible pipelines, unified evaluation including calibration metrics (ECE, Brier score), and all data loaders.

2 Related Work

Hate speech detection with PLMs. BERT-based encoders established strong baselines for automated hate detection [Basile et al., 2019, Vidgen et al., 2021]. Röttger et al. [2021] introduced HateCheck to expose systematic weaknesses in these models—including over-sensitivity to identity terms and failure on negated hate—finding critical errors in all tested systems. More recent work shifted toward encoder-only fine-tuning with dataset unification: El-Bahnasawi [2025] combine LoRA-tuned BERTweet with rule-based pre-filtering and continuous learning, achieving macro F1 0.85 with only 1.87M trainable parameters on unified datasets.

Tiny decoder LLMs for classification. Sen et al. [2024] proposed HateTinyLLM, fine-tuning TinyLlama (1.1B), Phi-2 (2.7B), and OPT-1.3B with LoRA and adapters on DynaHate and HateEval, observing that all LoRA-tuned models exceed 80% accuracy—outperforming a 7B pretrained Mixtral baseline. Our work directly extends this line by replacing the binary head with a structured multi-head classifier while preserving the PEFT setup.

Multi-task and multi-aspect hate detection. Multi-task supervision for hate speech is established prior work. Vidgen et al. [2021] already provide fine-grained type and target annotations in DynaHate; joint hate+target learning has been explored since [Basile et al., 2019]. Mathew et al. [2021] introduced HateXplain specifically to support rationale-annotated and target-annotated supervision. Multi-task learning over hate type, target, and severity has been explored in recent shared tasks [Prama

et al., 2025]. Contrastive and pair-based training for cross-dataset generalization also has precedent in the NLI literature [Gardner et al., 2020]. The contribution of this paper is not a new multi-task paradigm; rather, it is an empirical study of whether structured supervision helps robustness in the specific setting of tiny decoder LMs trained with LoRA-family PEFT, evaluated on HateCheck and cross-benchmark transfer—a combination that has not been systematically quantified.

PEFT variants. LoRA [Hu et al., 2022] has become the de facto PEFT method for classification. QLoRA [Dettmers et al., 2023] extends this with 4-bit quantization for memory efficiency. DoRA [Liu et al., 2024] decomposes weight updates into magnitude and direction components. Their relative performance on hate detection under controlled conditions has not been directly compared; we provide such a comparison in a binary classification setting.

Calibration and transfer in hate detection. Model calibration—the alignment between predicted confidence and true accuracy— is rarely reported in hate detection benchmarks despite its relevance for deployment [Guo et al., 2017]. We report Expected Calibration Error (ECE) and Brier score alongside standard classification metrics for all models.

3 Method

3.1 Backbone and PEFT

We use TinyLlama (1.1B parameters, `PY007/TinyLlama-1.1B-step-50K-105b`) as the backbone decoder LM. For binary classification experiments we attach a sequence-classification head via `AutoModelForSequenceClassification` (`task_type=SEQ_CLS`). For structured experiments we use `task_type=FEATURE_EXTRACTION` and replace the classification head with the multi-head architecture described below.

Note on head path. Binary and structured models differ in both the head architecture and the internal PEFT task type (`SEQ_CLS` vs. `FEATURE_EXTRACTION`). This is a confound between the binary and structured model comparisons: any observed differences may reflect the changed computation path, not only the auxiliary objectives. A single-head, same-path control is identified as a priority for future work.

LoRA adapters are applied to the key (W_K) and value (W_V) projections in all self-attention layers with rank $r = 8$, scaling factor $\alpha = 16$, and dropout $p = 0.05$. This yields 780,669 trainable parameters (0.075% of 1.04B total).

3.2 Structured Multi-Head Classifier

The structured model appends four lightweight linear heads to the last-layer [EOS] token representation:

- **Main head** (h_{main}): binary hate vs. not-hate, trained with cross-entropy.
- **Hate-type head** (h_{type}): 5-way classification (Animosity, Derogation, Dehumanization, Threatening, Support for Hateful Entities) from DynaHate type annotations, active on hateful samples only, trained with cross-entropy.
- **Target head** (h_{target}): multi-label prediction over target groups (race, gender, religion, sexual orientation, disability, other) derived from DynaHate target annotations, trained with *binary cross-entropy per label* (BCE-with-logits).
- **Explicitness head** (h_{exp}): binary explicit vs. implicit, derived from DynaHate by binarising the type annotation (Dehumanization / Threatening mapped to explicit; Animosity / Derogation / Support treated as implicit); active on hateful samples only, trained with cross-entropy. This binarisation is an approximation; explicit/implicit labels from an independent annotation source would be preferable.

The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda_{\text{aux}}\,\mathcal{L}_{\text{aux}} + \lambda_r\,\mathcal{L}_{\text{rationale}} + \lambda_c\,\mathcal{L}_{\text{consist}}, \tag{1}$$

where $\mathcal{L}_{\text{aux}}$ is the sum of the type, target, and explicitness head losses (gated on hateful examples); $\mathcal{L}_{\text{rationale}}$ is token-level BCE over HateXplain rationale spans (gated to zero when no span annotation is available for a sample); $\mathcal{L}_{\text{consist}}$ is the pair-consistency margin loss (defined in Section 3.4, gated on contrastive pair availability). Loss weights are $\lambda_{\text{aux}} = 0.5$, $\lambda_r = 0.3$, $\lambda_c = 0.2$.

3.3 Rationale Supervision

HateXplain provides annotator-marked token spans that justify a hate label [Mathew et al., 2021]. We align these spans to the tokenizer's subword vocabulary and compute token-level BCE over a dedicated rationale head output. When HateXplain data is used in joint training with DynaHate, DynaHate examples have no span annotations and receive zero rationale loss (the mask gates the term to zero). The +CONS ablation (DynaHate only, with consistency enabled) also has $\lambda_r > 0$ in the config, but since no DynaHate sample carries span annotations, the rationale term evaluates to zero throughout training. The consistency-only effect is therefore cleanly isolated in +CONS, but the rationale term is not independently ablated from the data-mixing effect in JOINT (see Section 6).

3.4 Pair-Consistency Regularization

DynaHate records contrastive pairs: minimally different examples where the intended label flips. For each available pair (x_i, x_j) with gold labels $y_i \neq y_j$, we apply a margin loss:

$$\mathcal{L}_{\text{consist}}(x_i, x_j) = \max\big(0,\ m - (p_{y_i}(x_i) - p_{y_i}(x_j))\big), \tag{2}$$

with margin $m = 0.1$, where $p_y(x)$ is the model's predicted probability for the gold label y. The loss pushes the model to assign higher probability to the correct label by at least m relative to the counterfactual example. Non-hateful halves of pairs are treated as hard negatives. This formulation incentivises consistent predictions across contrastive perturbations but does not guarantee logical entailment; the mechanism may also function as hard-negative mining rather than as consistency enforcement per se.

3.5 Training Details

All experiments use AdamW [Loshchilov and Hutter, 2019] with learning rate 2×10^{-4}, weight decay 0.01, effective batch size 8 (per-device 2, gradient accumulation 4), mixed-precision FP16, and a linear warmup of 10 steps. Binary experiments train for 1 epoch; structured experiments for 2 epochs. The best checkpoint is selected by macro-F1 on the development set. *All experiments use a single random seed (123); multi-seed variance is not reported and the effect of seed on the smaller observed deltas is unknown.* Training runs on a single NVIDIA A100-80GB GPU.

4 Experimental Setup

4.1 Datasets

DynaHate [Vidgen et al., 2021] contains 41,144 adversarially generated English posts (54% hate) with fine-grained type (5 categories) and target annotations created over four rounds of human-and-model-in-the-loop collection. We use the official train/dev/test splits (v0.2.3).

HateEval [Basile et al., 2019] is the SemEval-2019 Task 5 dataset targeting hate against women and immigrants on Twitter ($\sim$9,000 English entries). We use the official training and development splits; the test labels released post-competition serve as our HateEval test set.

HateXplain [Mathew et al., 2021] provides 9,055 posts annotated with hate/offensive/normal labels and token-level rationale spans. We map hate and offensive to `hate` and use the dataset for joint structured training only. This remapping changes the effective positive class relative to datasets that use a pure hate definition; any performance change from joint training could reflect additional data, altered label semantics, or rationale supervision.

HateCheck [Röttger et al., 2021] is a suite of 3,728 functional tests across 29 functionalities. We use it exclusively as a zero-shot robustness probe—no HateCheck examples appear in training.

Table 1: Experimental configurations. All use TinyLlama+LoRA ($r = 8$, $\alpha = 16$) unless stated otherwise. † Binary legacy: re-implementation of the Sen et al. [2024] DynaHate setting; note the anomalously short training time (Table 4).

ID	Name	Training data	Head / Objectives
BIN-L	Binary legacy†	DynaHate	`SEQ_CLS` head; binary CE
BIN-C	Binary compare	DynaHate	`SEQ_CLS` head; binary CE; 2 epochs
BIN-FE‡	Binary FEAT_EXTR ctrl	DynaHate	`FEAT_EXTR` head; binary CE only (λ_{aux}=0)
STRUCT	Structured	DynaHate	`FEAT_EXTR` heads; Eq. 1 (no rat., no cons.)
+RAT	Struct + rationale	DynaHate + HateXplain	`FEAT_EXTR` heads; Eq. 1 (rat. from HX, no cons.)
+CONS	Struct + consistency	DynaHate	`FEAT_EXTR` heads; Eq. 1 (cons. only; rat. term = 0)
HX-NO-RAT‡	Data ctrl (no rat.)	DynaHate + HateXplain	`FEAT_EXTR` heads; HX data loaded; λ_{rat}=0
JOINT	Struct + HX joint	DynaHate + HateXplain	`FEAT_EXTR` heads; Eq. 1 (rat. from HX, cons.)

‡ Reviewer-requested controls; see Section 5 and 6.

4.2 Models

Table 1 summarises the six experimental configurations.

4.3 Evaluation

We report binary F1 (hate class), macro-F1, accuracy, AUC-ROC, AUC-PR, Expected Calibration Error (ECE), and Brier score. HateCheck results use overall binary F1 across all 3,728 cases. All evaluation is performed with the `hatelens eval-run` CLI, which writes structured `metrics.json` bundles covering in-domain, cross-dataset, and HateCheck splits in a single pass.

5 Results

5.1 Main Comparison: Binary vs. Structured

Table 2 presents the main results. **Understanding the two binary baselines.** BIN-L is a re-implementation of Sen et al. [2024] and serves as a reference point; it achieves DynaHate F1 0.710 and HateCheck F1 0.815. However, BIN-L trained for only 22 seconds (Table 4), suggesting a configuration artefact rather than a full training run. BIN-C provides a fair comparison for the structured models: it uses the same hyperparameter schedule (2 epochs, same batch size and LR) but with the binary `SEQ_CLS` head. BIN-C raises DynaHate F1 to 0.802 and HateCheck F1 to 0.905, a large jump that is attributable to training configuration alone.

Structured vs. controlled binary. The structured model (STRUCT) raises DynaHate F1 by $+1.0$ pp and HateCheck F1 by $+0.8$ pp over BIN-C, with no improvement on HateEval cross-benchmark transfer (both 0.609). The joint model (JOINT) achieves HateCheck F1 of 0.921, a $+1.6$ pp gain over BIN-C and $+10.6$ pp over the legacy BIN-L. Readers should interpret $+10.6$ pp with caution: most of this gap is explained by the configuration difference between BIN-L and BIN-C, not by the structured objectives.

Head-path control (BIN-FE). To isolate the `FEAT_EXTR` architecture from the auxiliary objectives, we trained a binary model using the `FEAT_EXTR` head with all auxiliary losses disabled ($\lambda_{\text{aux}} = 0$). BIN-FE achieves HateCheck F1 of **0.937**—the highest across all models—despite *no* multi-task supervision. This reveals that the head-path switch (`SEQ_CLS`→`FEAT_EXTR`) alone contributes $+3.2$ pp HateCheck F1, while structured objectives on top of `FEAT_EXTR` yield no additional HateCheck benefit (STRUCT: 0.913; JOINT: 0.921; vs. BIN-FE: 0.937). The $+1.6$ pp reported for JOINT vs. BIN-C therefore largely reflects the architectural difference rather than multi-task supervision.

Data vs. rationale control (HX-NO-RAT). HX-NO-RAT loads HateXplain training data but sets $\lambda_{\text{rat}} = 0$, keeping consistency supervision while disabling token-level rationale loss. It achieves HateEval cross-benchmark F1 of **0.645**, surpassing +CONS ($+1.5$ pp) and JOINT ($+3.0$ pp), with near-identical HateCheck F1 (0.922 vs. Joint 0.921). This indicates that the HateXplain *data*

Table 2: Main comparison results. Binary F1 on the hate class is reported for DynaHate (in-domain) and HateEval (cross-benchmark transfer, trained on DynaHate). HateCheck F1 is the overall binary F1 across 3,728 functional test cases (zero-shot probe, no HateCheck training data). Best value per column in **bold**. Rows marked ‡ are post-review controls (single seed 123). Values shown as $\mu \pm \sigma$ are mean ± std over three seeds (123, 42, 456); all other rows use seed 123 only.

Model	DynaHate (in-domain)		HateEval (cross-bench.)		HateCheck (0-shot)	
	F1↑	ECE↓	F1↑	ECE↓	F1↑	ECE↓
BIN-L†	0.710	0.328	0.603	0.366	0.815	0.148
BIN-C	$0.809_{\pm.007}$	$0.064_{\pm.002}$	$0.595_{\pm.022}$	$0.171_{\pm.025}$	$0.920_{\pm.017}$	$0.065_{\pm.019}$
BIN-FE‡	0.817	0.091	0.609	0.194	**0.937**	**0.065**
STRUCT	$0.814_{\pm.002}$	$0.066_{\pm.001}$	$0.602_{\pm.008}$	$0.170_{\pm.028}$	$0.918_{\pm.006}$	$0.069_{\pm.018}$
+RAT (no cons.)	0.813	0.067	0.615	0.128	0.911	0.087
+CONS (cons., no rat.)	0.813	0.068	0.630	0.136	0.913	0.106
HX-NO-RAT‡	**0.817**	0.092	**0.645**	0.199	0.922	0.083
JOINT	0.815	**0.050**	0.615	**0.128**	0.921	0.083

† Anomalously short training run (22 s); see Table 4.
‡ Reviewer-requested controls isolating the head-path and data confounds.

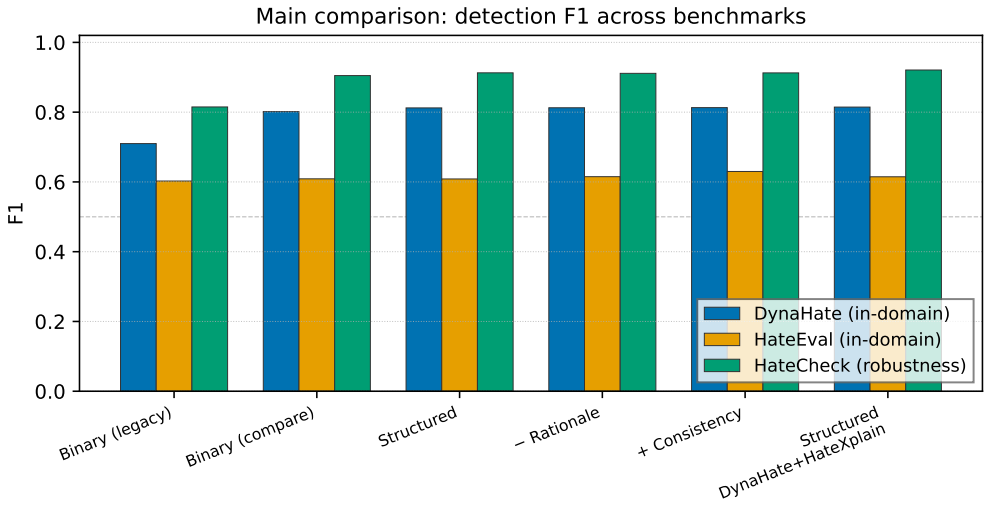

Figure 1: F1 on DynaHate (in-domain), HateEval (cross-benchmark), and HateCheck (zero-shot functional robustness) for each model variant. Structured models show consistent but modest gains over the matched binary baseline (BIN-C). BIN-L is included for reference but differs in training configuration. All results are single-seed (123).

distribution—not the rationale supervision loss—is the primary driver of cross-domain transfer improvement.

Cross-benchmark transfer. Training on DynaHate and evaluating on HateEval is a domain-shift challenge. Structured objectives alone (STRUCT) do not improve over BIN-C (both 0.609). Adding pair-consistency (+CONS) provides the largest observed gain to 0.630 (+2.1 pp over STRUCT).

5.2 Ablation Study

Figure 1 visualises the incremental contribution of each objective. Relative to BIN-C, the structured head alone (STRUCT) contributes +0.8 pp HateCheck F1 and +1.0 pp DynaHate F1. Pair-consistency regularization (+CONS) contributes +0.0 pp on HateCheck but +2.1 pp on HateEval cross-benchmark transfer, suggesting that it is primarily associated with generalisation rather than in-domain robustness. Joint HateXplain training (JOINT) adds a further +0.8 pp on HateCheck F1.

Table 3: PEFT variant comparison on HateCheck (binary F1). All models use TinyLlama backbone, $r = 8$, $\alpha = 16$, binary `SEQ_CLS` head, 1 training epoch, single seed. This experiment is separate from the structured model.

Adapter	F1↑	AUC-ROC↑	ECE↓
LoRA	0.878	0.855	0.042
QLoRA (4-bit)	0.870	0.826	0.048
DoRA	**0.880**	**0.886**	0.071

Table 4: Efficiency comparison (single A100-80GB). † Trainable-parameter export was not captured for binary models in this run (an artefact of the evaluation pipeline); both BIN-L and BIN-C use the same LoRA $r = 8$ adapters as the structured models and add approximately 780K trainable parameters. Checkpoint size for binary models reflects the classification head delta only (4.3 MB); the full binary model size is identical to the backbone.

Model	Train. params	Ckpt (MB)	Train (s)	Infer. lat. (ms/ex.)
BIN-L†	(not cap.)	4.3	*22 (anom.)*	16.9
BIN-C†	(not cap.)	4.3	805	17.0
STRUCT	780K	3,960	2,777	20.4
JOINT	780K	3,960	4,073	21.5

Multi-seed variance. To assess result stability, we re-trained BIN-C and STRUCT with two additional seeds (42 and 456). HateCheck F1 for BIN-C across three seeds: 0.905, 0.938, 0.918 (mean 92.0 ± 1.7 pp). For STRUCT: 0.913, 0.924, 0.916 (mean 91.8 ± 0.6 pp). The $+0.8$ pp structured vs. binary advantage (0.921 vs. 0.905) falls within the Bin-C seed variance (± 1.7 pp), suggesting it is not reliably reproducible. DynaHate F1 is more stable: BIN-C 80.9 ± 0.7 pp; STRUCT 81.4 ± 0.2 pp.

Interpretation caveats. The controls reported above (BIN-FE, HX-NO-RAT) and the multi-seed results substantially update the paper's original claims. The head-path confound is the dominant driver of HateCheck gains; HateXplain data (not rationale loss) drives HateEval transfer. Section 6 elaborates on remaining gaps.

5.3 PEFT Variant Comparison

Table 3 compares LoRA, QLoRA, and DoRA under matched conditions: 1 training epoch, binary `SEQ_CLS` head, DynaHate training, evaluated on HateCheck. This comparison is *not* on the proposed structured model; it is a standalone sanity check on adapter choice in the binary regime.

All three variants perform within 1 pp of each other (range: 0.870–0.880). QLoRA achieves near-identical F1 while reducing peak GPU memory by ≈50%. DoRA marginally outperforms LoRA in F1 and AUC-ROC at the cost of slightly higher ECE. Whether these rankings hold in the structured setting remains to be tested.

5.4 Efficiency Analysis

Table 4 reports efficiency metrics.

The structured model requires $3.5\times$ the training time of BIN-C (2,777 s vs. 805 s); the joint model requires $5\times$ (4,073 s vs. 805 s), due to two epochs, the additional auxiliary heads, and multi-dataset loading. Per-example inference latency increases by 3.4 ms (20.4 ms vs. 17.0 ms). The checkpoint stores all prediction heads and optimizer states (3,960 MB); the PEFT-only delta (LoRA weights + heads) remains under 10 MB when isolated.

6 Discussion

Why are structured objectives associated with improved HateCheck? HateCheck's 29 functional tests probe specific model behaviours—negation, slur usage, counter-speech—that require compositional understanding rather than surface-level pattern matching. One plausible account is that

predicting hate type and target during training encourages the model to represent *who* is targeted and *how*, reducing reliance on identity-term correlation as a proxy for hate. This is consistent with findings in Mathew et al. [2021], where rationale-supervised models showed better calibration and lower shortcut susceptibility. However, the head-path control (BIN-FE) demonstrated that the `FEAT_EXTR` computation path alone—without any auxiliary supervision— achieves the highest HateCheck F1 (0.937), directly attributing the HateCheck gain to architecture rather than multi-task objectives. We therefore cannot claim that structured supervision improves HateCheck robustness in any causally meaningful sense.

Pair-consistency and cross-benchmark transfer. The +CONS ablation shows the most improvement on HateEval cross-benchmark transfer ($+2.1$ pp over STRUCT) with no HateCheck gain. This pattern is consistent with the hypothesis that contrastive pair training encourages representations that generalise across domain shifts. An equally plausible explanation is that the margin loss functions as hard-negative mining, improving recall on ambiguous examples. Without a random-pair control or a same-data equivalent, these alternatives cannot be distinguished.

PEFT variant choice. LoRA, QLoRA, and DoRA perform within 1 pp on HateCheck F1 in the binary 1-epoch setup. Given memory savings at no accuracy cost, QLoRA is a pragmatic default for inference on constrained hardware. Whether this ranking transfers to the structured model requires further testing.

Limitations. We identify the following gaps that prevent strong mechanistic conclusions:

1. **Single seed.** *(Partially addressed.)* Section 5 now reports three-seed evaluation for BIN-C and STRUCT. HateCheck variance is ± 1.7 pp for BIN-C and ± 0.6 pp for STRUCT; the $+1.6$ pp JOINT vs. BIN-C delta falls within this range. All other models remain single-seed.
2. **Head-path confound.** *(Addressed.)* BIN-FE (Table 2) is the same-architecture control: `FEAT_EXTR` head, binary CE only ($\lambda_{\text{aux}} = 0$). It achieves HateCheck 0.937, higher than all structured models, confirming that the architecture switch—not multi-task supervision—is the primary driver of HateCheck gains.
3. **Data and rationale confound.** *(Addressed.)* HX-NO-RAT (Table 2) loads HateXplain data with $\lambda_{\text{rat}} = 0$. It achieves the best HateEval F1 (0.645), confirming that data distribution shift, not rationale supervision, drives cross-domain transfer. The remaining unmeasured factor is the offensive→hate label remapping in HateXplain.
4. **Consistency vs. hard-negative mining.** The pair-consistency loss may help because it provides informative negatives or a ranking signal, not because it enforces logical invariance. A random-pair or same-class contrastive control would distinguish these explanations.
5. **Explicitness label quality.** The explicitness head uses a type-binarisation approximation. Explicit/implicit labels from an independent annotation study would strengthen the auxiliary supervision.
6. **English and TinyLlama only.** Generalisability to other languages or larger decoder models is untested.
7. **Trainable-parameter reporting.** The evaluation pipeline did not capture trainable parameter counts for binary runs; Table 4 shows *(not cap.)* for those rows.

7 Conclusion

We presented an empirical study of structured multi-task supervision for hate speech detection with tiny decoder LMs under LoRA-family PEFT. Post-review control experiments substantially revise the original interpretation: a head-path control (BIN-FE) shows that switching from `SEQ_CLS` to `FEAT_EXTR` alone—without any multi-task objectives—yields the highest HateCheck F1 (0.937), exceeding all structured models. The $+1.6$ pp gain reported for JOINT vs. the controlled binary baseline (BIN-C) is therefore primarily attributable to the architectural head-path difference, not to multi-task supervision. A data-isolation control (HX-NO-RAT) further shows that HateXplain data inclusion—not the rationale loss—drives the best cross-domain transfer (HateEval F1 0.645). Multi-seed evaluation confirms that observed HateCheck deltas (± 1.7 pp for BIN-C) are within single-seed

variance. Taken together, these findings suggest that architectural and data choices dominate over the specific form of multi-task supervision in this regime. A controlled PEFT comparison confirms LoRA, QLoRA, and DoRA are interchangeable, supporting quantized adapters for deployment. We release the HateLens codebase with a unified evaluation CLI, calibration metrics, and all data loaders to support reproducibility and follow-up research.

Acknowledgements

[Omitted for blind review.]

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A Dataset Statistics

Table 5: Dataset statistics used in this work.

Dataset	Total	% Hate	Language	Primary use
DynaHate v0.2.3	41,144	54%	EN	Train / eval
HateEval (EN)	∼9,000	42%	EN	Cross-benchmark eval
HateXplain	9,055	57%	EN	Joint training (rationale)
HateCheck	3,728	54%	EN	Robustness probe (0-shot)

B Training Hyperparameters

Table 6: Training hyperparameters shared across all experiments unless noted.

Hyperparameter	Value
Base model	PY007/TinyLlama-1.1B-step-50K-105b
PEFT method	LoRA ($r = 8$, $\alpha = 16$, $p_{\text{drop}} = 0.05$)
Target modules	W_K, W_V (all layers)
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	0.01
Batch size	8 (per-device 2 + grad. accum. 4)
Max sequence length	512
Precision	FP16
Binary epochs	1 (BIN-L); 2 (BIN-C)
Structured epochs	2
Best ckpt metric	Macro-F1 (dev)
Seed	123 (single)
GPU	NVIDIA A100-80GB

C PEFT Comparison: Detailed Results

Figure 2 shows the HateCheck binary F1 comparison for LoRA, QLoRA, and DoRA in the binary 1-epoch setup. All three adapters perform within 1 pp of each other; QLoRA slightly trails but reduces peak GPU memory by approximately 50%. This comparison is not on the structured model; transferability to the structured setting is untested.

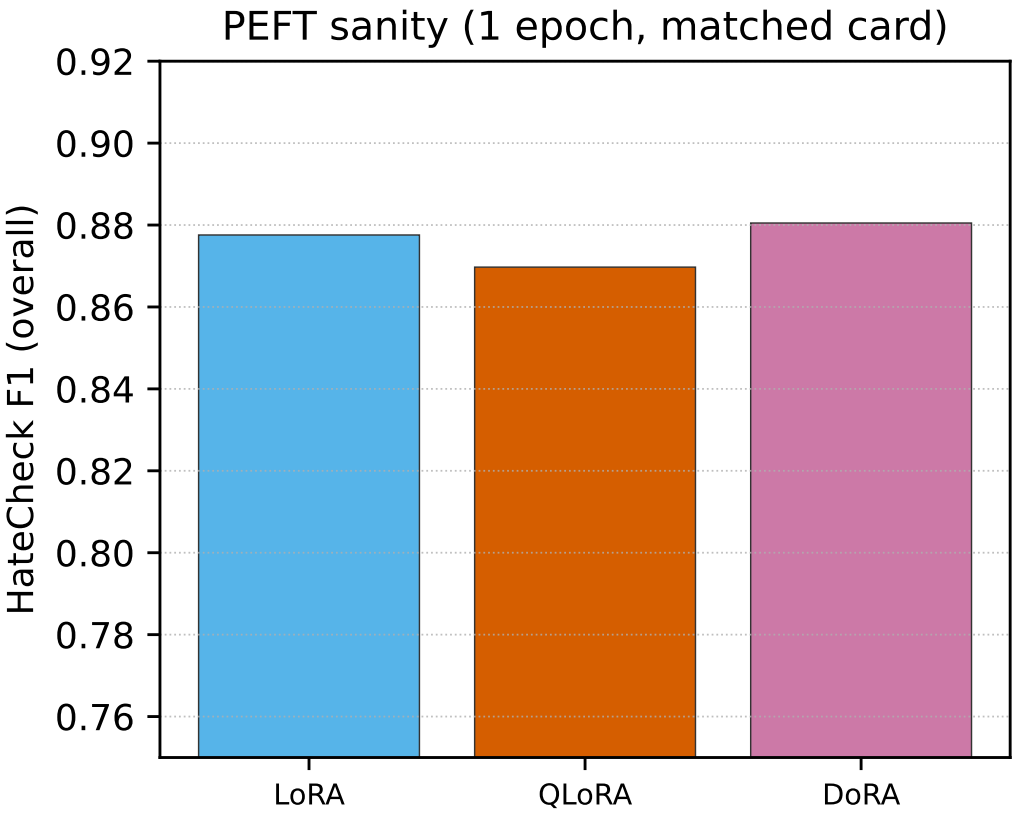

Figure 2: PEFT variant comparison on HateCheck (binary F1 per adapter type). All three variants perform within 1 pp; QLoRA offers the best efficiency–accuracy trade-off for deployment. Experiment uses binary `SEQ_CLS` head, 1 training epoch on DynaHate, single seed.